

Slope & Rate of Change Answer Key

Grade 7 · Trail Ranger Academy · Tutor copy

What is in this kit

- **Classwork 1 – Summit Signal** (44 Qs): vocabulary, types of slope, slope from graphs (incl. scaled axes), trail-map segments, steeper vs flatter, plotting from a slope.
- **Classwork 2 – Glacier Express** (49 Qs): slope from two points, tables, real-world rates with units, comparing rates, horizontal/vertical lines, missing coordinates, error analysis.
- **Quiz 1 – Weather Balloon Roundup** and **Quiz 2 – Slalom Run** (12 Qs each, no briefing). **Homework** (14 decoder locks + 6 written questions) and **Class Notes** (one page).

Scoring rules in the games: +100 first try, +50 second try (hint shows after a miss, answer revealed after two misses), +25 speed bonus, golden question ×2, Boost ×2, 3-in-a-row = free power-up, 5-in-a-row = +100. An answer that has the right value but is not simplified is *not* counted as a miss: the game nudges the student to simplify. Use ⚙️ (top right) to jump to any camp/question or show an answer.

Suggested use

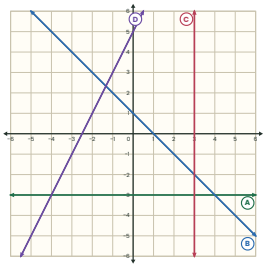
- Classwork 1 ≈ 60–75 min (or split: camps 1–3, then 4–5 + bonus). Classwork 2 ≈ 70–85 min.
- Quiz 1 after Classwork 1, Quiz 2 after Classwork 2 (about 15 min each). Homework after both classworks.
- The games do not save progress. Use the ⚙️ jump menu to resume a split lesson.

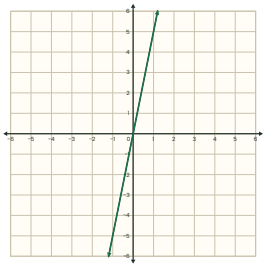
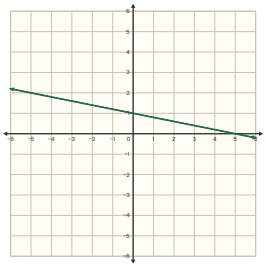
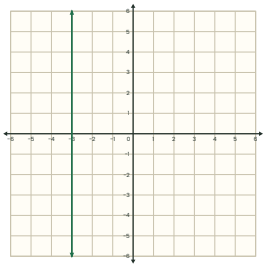
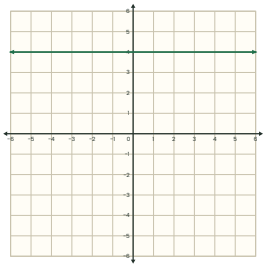
Summit Signal – Classwork 1

⛺ Base Camp — Slope Vocabulary

#	QUESTION	ANSWER	WHY
1.1	The vertical change between two points on a graph is called the _____.	**Rise**	Vertical (up/down) change is the **rise**.
1.2	The horizontal change between two points is called the _____.	**Run**	Horizontal (left/right) change is the **run**.
1.3	Slope is the _____ of change in a relationship.	**rate**	Slope is the **rate** of change: how fast y changes as x changes.
1.4	Slope is a ratio: change in the _____ variable over change in the _____ variable.	**dependent ÷ independent**	Slope = (change in the **dependent** variable y) ÷ (change in the **independent** variable x).
1.5	On the graph of a line, slope describes the _____ of the line.	**steepness**	Slope measures how **steep** a line is.
1.6	Which formula gives the slope of a line?	**rise ÷ run**	Slope = **rise ÷ run** (vertical change over horizontal change).
1.7	A hiker walks 4 m across and climbs 12 m up. Fill in the rise, the run, and the slope.	**rise = 12, run = 4, slope = 3**	Rise = 12, run = 4. Slope = 12 ÷ 4 = **3**.

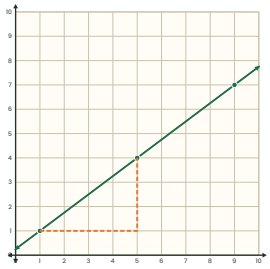
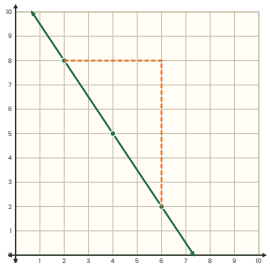
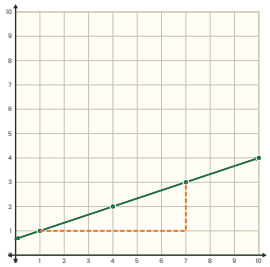
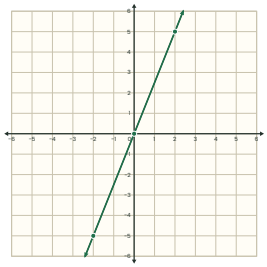
🌲 Pine Ridge — Types of Slope

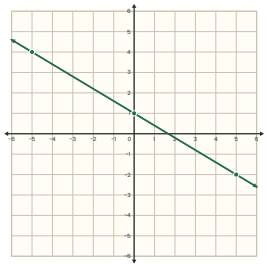
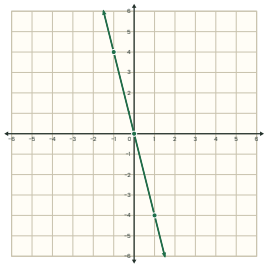
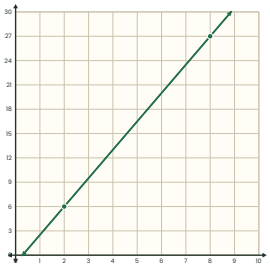
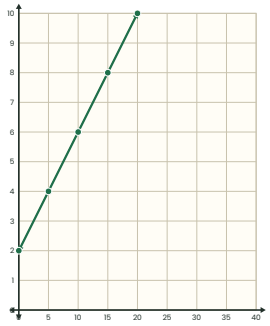
#	QUESTION	ANSWER	WHY
2.1	Look at each line from left to right. What type of slope does each line have? 	**Line A: Zero; Line B: Negative; Line C: Undefined; Line D: Positive**	Positive lines climb, negative lines slide down, horizontal lines are zero, vertical lines are undefined.
2.2	A graph with a _____ slope goes up from left to right.	**Positive**	Going up from left to right = **positive** slope.
2.3	The slope of any horizontal line is always _____.	**Zero**	A horizontal line has rise = 0, so slope = 0 ÷ run = **0**.
2.4	The slope of any vertical line is always _____.	**Undefined**	A vertical line has run = 0, and dividing by 0 is impossible, so the slope is **undefined**.
2.5	Match each trail to its type of slope.	**A wheelchair ramp going up: Positive; A ski run going downhill: Negative; A flat lakeside path: Zero; A straight cliff wall: Undefined**	The ramp climbs (positive), the ski run drops (negative), the lake path is flat (zero), the cliff wall is vertical (undefined).

2.6	Which memory trick is correct?	**Undefined = Up-and-down (vertical). Zero = flat (horizontal).**	**U**ndefined → **U**p-and-down (vertical). **Z**ero → flat like the floor (horizontal).
2.7	What type of slope does this line have? 	**Positive**	Reading left to right, the line goes **up**, so the slope is **positive**.
2.8	What type of slope does this line have? 	**Negative**	Reading left to right, the line goes **down**, so the slope is **negative**.
2.9	What type of slope does this line have? 	**Undefined**	The line is **straight up and down** (vertical): the run is 0 and you can't divide by 0, so the slope is **undefined**.
2.10	What type of slope does this line have? 	**Zero**	The line is **flat** (horizontal): the rise is 0, so the slope is **zero**.

🪜 Staircase Pass — Count the Steps

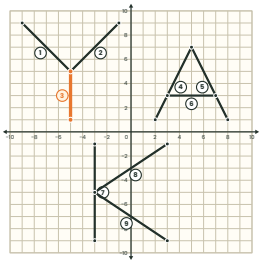
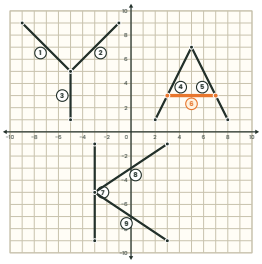
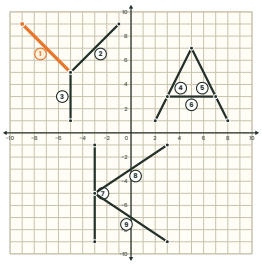
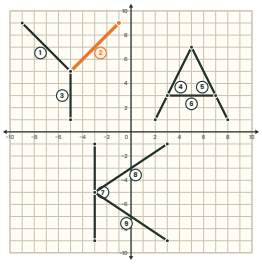
#	QUESTION	ANSWER	WHY

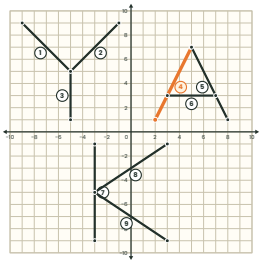
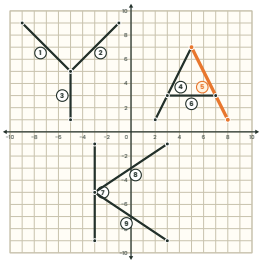
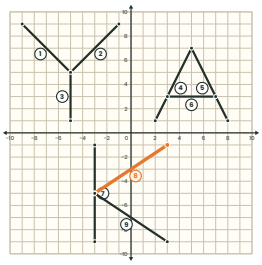
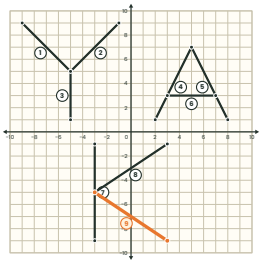
3.1	Use the dashed staircase. Count the rise (up is positive, down is negative) and the run, then write the slope. 	**rise = 3, run = 4, slope = 3/4**	On the graph: Rise = $y_2 - y_1 = 4 - 1 = 3$. Run = $x_2 - x_1 = 5 - 1 = 4$. Slope = rise ÷ run = **3/4**.
3.2	Use the dashed staircase. Count the rise (up is positive, down is negative) and the run, then write the slope. 	**rise = -6, run = 4, slope = -3/2**	On the graph: Rise = $y_2 - y_1 = 2 - 8 = -6$. Run = $x_2 - x_1 = 6 - 2 = 4$. Slope = $-6/4$ = **-3/2** (simplified).
3.3	Use the dashed staircase. Count the rise (up is positive, down is negative) and the run, then write the slope. 	**rise = 2, run = 6, slope = 1/3**	On the graph: Rise = $y_2 - y_1 = 3 - 1 = 2$. Run = $x_2 - x_1 = 7 - 1 = 6$. Slope = $2/6$ = **1/3** (simplified).
3.4	Find the slope of the line. Whole number or simplified fraction. 	**5/2**	On the graph: Rise = $y_2 - y_1 = 5 - (-5) = 10$. Run = $x_2 - x_1 = 2 - (-2) = 4$. Slope = $10/4$ = **5/2** (simplified).

3.5	Find the slope of the line. Whole number or simplified fraction. 	**-3/5**	On the graph: Rise = $y_2 - y_1 = (-2) - 4 = -6$. Run = $x_2 - x_1 = 5 - (-5) = 10$. Slope = -6/10 = **-3/5** (simplified).
3.6	Find the slope of the line. Whole number or simplified fraction. 	**-4**	On the graph: Rise = $y_2 - y_1 = (-4) - 4 = -8$. Run = $x_2 - x_1 = 1 - (-1) = 2$. Slope = -8/2 = **-4** (simplified).
3.7	Find the slope of the line. Careful: each square going up the y-axis is worth 3. Whole number or simplified fraction. 	**7/2**	On the graph: Rise = $y_2 - y_1 = 27 - 6 = 21$. Run = $x_2 - x_1 = 8 - 2 = 6$. Slope = 21/6 = **7/2** (simplified).
3.8	Find the slope of the line. Careful: each square going across the x-axis is worth 5. Whole number or simplified fract… 	**2/5**	On the graph: Rise = $y_2 - y_1 = 10 - 2 = 8$. Run = $x_2 - x_1 = 20 - 0 = 20$. Slope = 8/20 = **2/5** (simplified).

🗺️ Yak Ridge Map Room — Trail Map Segments

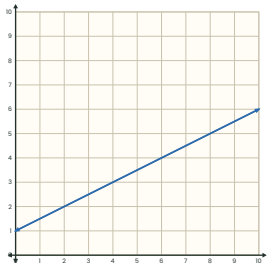
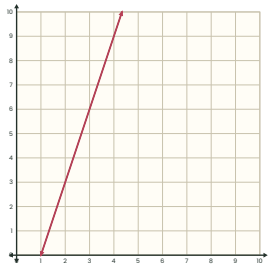
#	QUESTION	ANSWER	WHY

4.1	Find the slope of segment 3 (the thick orange one). Whole number, simplified fraction, or undefined. 	**undefined**	Segment 3: Both points have x = -5, so the run is (-5) - (-5) = 0. You can't divide by zero, so the slope is **undefined** (a vertical line).
4.2	Find the slope of segment 6 (the thick orange one). Whole number, simplified fraction, or undefined. 	**0**	Segment 6: Both points have y = 3, so the rise is 0. 0 ÷ 4 = **0** (a horizontal line).
4.3	Find the slope of segment 1 (the thick orange one). Whole number, simplified fraction, or undefined. 	**-1**	Segment 1: Rise = $y_2 - y_1 = 5 - 9 = -4$. Run = $x_2 - x_1 = (-5) - (-9) = 4$. Slope = -4/4 = **-1** (simplified).
4.4	Find the slope of segment 2 (the thick orange one). Whole number, simplified fraction, or undefined. 	**1**	Segment 2: Rise = $y_2 - y_1 = 9 - 5 = 4$. Run = $x_2 - x_1 = (-1) - (-5) = 4$. Slope = 4/4 = **1** (simplified).

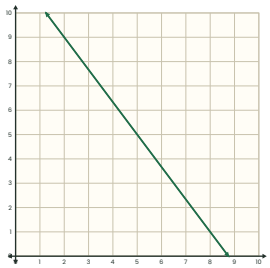
4.5	Find the slope of segment 4 (the thick orange one). Whole number, simplified fraction, or undefined. 	**2**	Segment 4: Rise = $y_2 - y_1 = 7 - 1 = 6$. Run = $x_2 - x_1 = 5 - 2 = 3$. Slope = 6/3 = **2** (simplified).
4.6	Find the slope of segment 5 (the thick orange one). Whole number, simplified fraction, or undefined. 	**-2**	Segment 5: Rise = $y_2 - y_1 = 1 - 7 = -6$. Run = $x_2 - x_1 = 8 - 5 = 3$. Slope = -6/3 = **-2** (simplified).
4.7	Find the slope of segment 8 (the thick orange one). Whole number, simplified fraction, or undefined. 	**2/3**	Segment 8: Rise = $y_2 - y_1 = (-1) - (-5) = 4$. Run = $x_2 - x_1 = 3 - (-3) = 6$. Slope = 4/6 = **2/3** (simplified).
4.8	Find the slope of segment 9 (the thick orange one). Whole number, simplified fraction, or undefined. 	**-2/3**	Segment 9: Rise = $y_2 - y_1 = (-9) - (-5) = -4$. Run = $x_2 - x_1 = 3 - (-3) = 6$. Slope = -4/6 = **-2/3** (simplified).

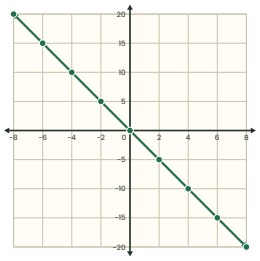
⛰️ Switchback Summit — Steeper or Flatter?

#	QUESTION	ANSWER	WHY

5.1	Which line is steeper? **Line A**  **Line B** 	**Line B**	Line A: slope 1/2 (rise 1 per 2 across). Line B: slope 3 (rise 3 per 1 across). 3 is bigger than 1/2, so **Line B** is steeper.
5.2	Which slope makes a steeper line: 2/3 or 5/3?	**5/3**	Both have a bottom of 3, and 5 > 2, so **5/3** is steeper.
5.3	Tap the slopes in order from the steepest line to the flattest line.	**3 → 5/2 → 1 → 1/2**	3 > 5/2 > 1 > 1/2, so the order is 3, 5/2, 1, 1/2.
5.4	Which line is steeper: one with slope -4 or one with slope 2?	**Slope -4**	Steepness depends on the size of the number ignoring the sign. |-4| = 4 is bigger than 2, so slope **-4** is steeper (it just goes downhill).
5.5	The line passes through the red point (-3, -2) and has slope 2/3. Tap a second point on the line.	**(0, 0) or (-6, -4)**	Slope 2/3 = rise 2 ÷ run 3. From (-3, -2) go 3 right and 2 up to **(0, 0)** (or the other way: **(-6, -4)**).
5.6	The line passes through the red point (-4, 3) and has slope -3/2. Tap a second point on the line.	**(-2, 0) or (-6, 6)**	Slope -3/2 = rise -3 ÷ run 2. From (-4, 3) go 2 right and 3 down to **(-2, 0)** (or the other way: **(-6, 6)**).

Bonus: Beacon Tower — Summit Challenge

#	QUESTION	ANSWER	WHY
6.1	Find the slope of the line. Whole number or simplified fraction. 	**-4/3**	On the graph: Rise = $y_2 - y_1 = 1 - 9 = -8$. Run = $x_2 - x_1 = 8 - 2 = 6$. Slope = -8/6 = **-4/3** (simplified).

6.2	Find the slope of the line. Careful: x counts by 2s and y counts by 5s. Whole number or simplified fraction. 	**-5/2**	On the graph: Rise = $y_2 - y_1 = (-10) - 10 = -20$. Run = $x_2 - x_1 = 4 - (-4) = 8$. Slope = -20/8 = **-5/2** (simplified).
6.3	The red point is (-2, 1). Tap a second point so the line through them has a slope of undefined.	**x = -2 (any point straight above/below)**	Undefined slope = vertical line, so x stays -2. Any point like (-2, y) works. For example **(-2, -6)**.
6.4	The red point is (3, -2). Tap a second point so the line through them has a slope of 0.	**y = -2 (any point straight left/right)**	Slope 0 = horizontal line, so y stays -2. Any point like (x, -2) works. For example **(-6, -2)**.
6.5	Which statement is TRUE?	**A line that goes down from left to right has a negative slope.**	Downhill = negative. Vertical = undefined, horizontal = 0, and slope is rise ÷ run.

Glacier Express – Classwork 2

📍 Two-Point Platform — Slope from Two Points

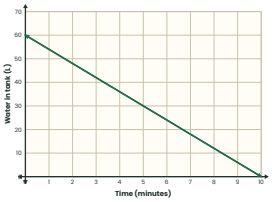
#	QUESTION	ANSWER	WHY
1.1	Which formula finds the slope between (x_1, y_1) and (x_2, y_2)?	**$(y_2 - y_1) \div (x_2 - x_1)$**	Rise = $y_2 - y_1$ and run = $x_2 - x_1$, so slope = **$(y_2 - y_1) \div (x_2 - x_1)$**.
1.2	Find the slope of the line through (2, 3) and (6, 11). Rise = $y_2 - y_1$ and run = $x_2 - x_1$, using (6, 11) as point 2.	**rise = 8, run = 4, slope = 2**	Rise = $y_2 - y_1$ = 11 - 3 = 8. Run = $x_2 - x_1$ = 6 - 2 = 4. Slope = 8/4 = **2** (simplified).
1.3	Find the slope of the line through (-1, 5) and (3, -1). Rise = $y_2 - y_1$ and run = $x_2 - x_1$, using (3, -1) as point 2.	**rise = -6, run = 4, slope = -3/2**	Rise = $y_2 - y_1$ = (-1) - 5 = -6. Run = $x_2 - x_1$ = 3 - (-1) = 4. Slope = -6/4 = **-3/2** (simplified).
1.4	Find the slope of the line through (-4, -2) and (2, 1). Rise = $y_2 - y_1$ and run = $x_2 - x_1$, using (2, 1) as point 2.	**rise = 3, run = 6, slope = 1/2**	Rise = $y_2 - y_1$ = 1 - (-2) = 3. Run = $x_2 - x_1$ = 2 - (-4) = 6. Slope = 3/6 = **1/2** (simplified).
1.5	Find the slope of the line through (0, -3) and (5, 1). Whole number or simplified fraction (or undefined).	**4/5**	Rise = $y_2 - y_1$ = 1 - (-3) = 4. Run = $x_2 - x_1$ = 5 - 0 = 5. Slope = rise ÷ run = **4/5**.
1.6	Find the slope of the line through (-6, 4) and (-2, -4). Whole number or simplified fraction (or undefined).	**-2**	Rise = $y_2 - y_1$ = (-4) - 4 = -8. Run = $x_2 - x_1$ = (-2) - (-6) = 4. Slope = -8/4 = **-2** (simplified).
1.7	Find the slope of the line through (7, -1) and (-2, 5). Whole number or simplified fraction (or undefined).	**-2/3**	Rise = $y_2 - y_1$ = 5 - (-1) = 6. Run = $x_2 - x_1$ = (-2) - 7 = -9. Slope = 6/-9 = **-2/3** (simplified).
1.8	Find the slope of the line through (-8, -3) and (4, 6). Whole number or simplified fraction (or undefined).	**3/4**	Rise = $y_2 - y_1$ = 6 - (-3) = 9. Run = $x_2 - x_1$ = 4 - (-8) = 12. Slope = 9/12 = **3/4** (simplified).
1.9	Jordan finds the slope through (1, 2) and (4, 8) but starts with the second point: (2 - 8) ÷ (1 - 4). Does the answer…	**No — it's still 2, as long as both subtractions use the same order**	(2 - 8) ÷ (1 - 4) = -6 ÷ -3 = **2**. Two negatives make a positive, so the order doesn't matter if you stay consistent.

📋 Table Tunnel — Slope from a Table

#	QUESTION	ANSWER	WHY
2.1	Find the slope (rate of change) shown in the table. Whole number or simplified fraction. X: 0, 2, 4, 6 Y: 3, 7, 11, 15	**2**	Pick two rows, like (0, 3) and (2, 7). Change in y = 7 - 3 = 4. Change in x = 2 - 0 = 2. Slope = 4/2 = **2**.
2.2	Find the slope (rate of change) shown in the table. Whole number or simplified fraction. X: 1, 2, 3, 4 Y: 10, 7, 4, 1	**-3**	Pick two rows, like (1, 10) and (2, 7). Change in y = 7 - 10 = -3. Change in x = 2 - 1 = 1. Slope = -3/1 = **-3**.

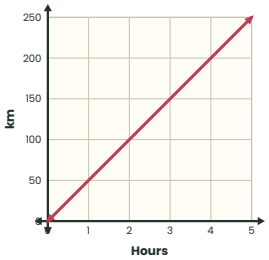
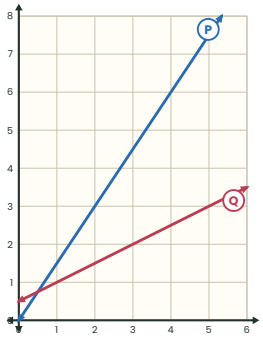
#			
2.3	Find the slope (rate of change) shown in the table. Whole number or simplified fraction. **X** 0 5 10 15 **Y** 2 5 8 11	**3/5**	Pick two rows, like (0, 2) and (5, 5). Change in y = 5 - 2 = 3. Change in x = 5 - 0 = 5. Slope = 3/5 = **3/5**.
2.4	This table shows a constant rate of change. Find the missing y-value (the ?). **X** 2 4 6 8 **Y** 5 9 **?** 17	**13**	The rate is constant. Use two known columns to find it: (2, 5) and (4, 9) give slope 2. From x = 4 to x = 6, y changes by 2 × 2. So the missing y is **13**.
2.5	Is the rate of change constant in this table? **X** 1 2 3 4 **Y** 2 5 9 14	**No — y goes up by 3, then 4, then 5**	x goes up by 1 each time, but y changes by +3, +4, +5. The rate isn't constant, so the points are **not** on one straight line.
2.6	Find the slope (rate of change) shown in the table. Whole number or simplified fraction. **X** -4 -2 0 2 **Y** 7 4 1 -2	**-3/2**	Pick two rows, like (-4, 7) and (-2, 4). Change in y = 4 - 7 = -3. Change in x = -2 - (-4) = 2. Slope = -3/2 = **-3/2**.
2.7	Find the slope (rate of change) shown in the table. Whole number or simplified fraction. **X** 1 4 6 10 **Y** 3 9 13 21	**2**	Pick two rows, like (1, 3) and (4, 9). Change in y = 9 - 3 = 6. Change in x = 4 - 1 = 3. Slope = 6/3 = **2**.

Rate Rail — Rates in the Real World

#	QUESTION	ANSWER	WHY
3.1	A hiker climbs from 1,200 m to 1,500 m in 30 minutes. What is the rate of change in meters per minute?	**10 m per min**	(1,500 - 1,200) ÷ 30 = 300 ÷ 30 = **10** meters per minute.
3.2	The graph shows the water left in a draining tank. What is the rate of change in liters per minute? 	**-6 L per min**	Rise = $y_2 - y_1 = 0 - 60 = -60$. Run = $x_2 - x_1 = 10 - 0 = 10$. Slope = -60/10 = **-6** (simplified). The rate is negative because the water is **decreasing**.
3.3	A ski lift rises 450 m in 5 minutes. What is its rate of change in meters per minute?	**90 m per min**	450 ÷ 5 = **90** meters per minute.
3.4	On a graph of snow depth over time, the slope is 12 cm per hour. What does that mean?	**The snow gets 12 cm deeper each hour**	A slope of 12 cm per hour means the snow depth **increases by 12 cm every hour**.
3.5	A candle burns from 20 cm tall to 14 cm tall in 3 hours. What is the rate of change in cm per hour?	**-2 cm per h**	(14 - 20) ÷ 3 = -6 ÷ 3 = **-2** cm per hour.

3.6	The table shows the cost of bags of trail mix. What is the unit rate (the slope) in dollars per bag? **BAGS** 2 4 6 8 **COST ($)** 6 12 18 24	**3 $ per bag**	Every 2 bags adds $6, so slope = 6/2 = **3** dollars per bag.
3.7	A drone climbs 15 m in 6 seconds. What is its rate of change in meters per second? Simplified fraction.	**5/2 m per s**	15/6 = **5/2** meters per second.
3.8	Which situation has a negative rate of change?	**A balloon sinking from 800 m down to 500 m**	The balloon's height goes **down** as time goes up, so its rate of change is negative.

⚖️ Compare Crossing — Compare the Rates

#	QUESTION	ANSWER	WHY
4.1	Which train is faster? **Train A** **HOURS** 1 2 3 **KM** 45 90 135 **Train B** 	**Train B**	Train A: 45 km per hour (table). Train B: 200 ÷ 4 = 50 km per hour (graph). 50 > 45, so **Train B** is faster.
4.2	Hiker 1 climbs 240 m in 8 minutes. Hiker 2's progress is in the table. Who climbs faster? **MINUTES** 0 5 10 **METERS** 0 125 250	**Hiker 1**	Hiker 1: 240 ÷ 8 = 30 m/min. Hiker 2: 125 ÷ 5 = 25 m/min. **Hiker 1** is faster.
4.3	Which line is steeper, P or Q? 	**Line P**	Line P goes through (0, 0) and (4, 6): slope 6/4 = 3/2. Line Q goes through (1, 1) and (5, 3): slope 2/4 = 1/2. **P** is steeper.
4.4	Tap these slopes from steepest to flattest.	**5/6 → 3/4 → 2/3 → 1/2**	In twelfths: 5/6 = 10/12, 3/4 = 9/12, 2/3 = 8/12, 1/2 = 6/12. So the order is 5/6, 3/4, 2/3, 1/2.
4.5	Bike A rides at 12 miles per hour. Bike B rides 100 miles in 8 hours. Which is faster?	**Bike B**	Bike B: 100 ÷ 8 = 12.5 mph, which is more than 12. **Bike B** is faster.
4.6	Line K passes through (0, 1) and (3, 7). The table shows line M. Do they have the same slope? **X** 1 2 3 **Y** 4 6 8	**Yes, both have slope 2**	Line K: (7 - 1) ÷ (3 - 0) = 2. Line M: y goes up by 2 each time x goes up by 1, so slope 2. **Same slope!**

🚧 Switchyard 5 — Flat, Wall & Missing Pieces

#	QUESTION	ANSWER	WHY
5.1	What is the slope of the line y = 5?	**Zero (0)**	**y = 5** is a horizontal line, so the slope is **0**.
5.2	What is the slope of the line x = -2?	**Undefined**	**x = -2** is a vertical line, so the slope is **undefined**.
5.3	Find the slope of the line through (3, -2) and (3, 6). Whole number or simplified fraction (or undefined).	**undefined**	Both points have x = 3, so the run is 3 - 3 = 0. You can't divide by zero, so the slope is **undefined** (a vertical line).
5.4	Find the slope of the line through (-5, 2) and (4, 2). Whole number or simplified fraction (or undefined).	**0**	Both points have y = 2, so the rise is 0. 0 ÷ 9 = **0** (a horizontal line).
5.5	The line passes through the red point (1, -2) and has slope -1/2. Tap a second point on the line.	**(3, -3) or (-1, -1)**	Slope -1/2 = rise -1 ÷ run 2. From (1, -2) go 2 right and 1 down to **(3, -3)** (or the other way: **(-1, -1)**).
5.6	The slope of the line through (1, 2) and (5, y) is 3/4. Find y. Type a whole number.	**5**	Run = 5 - 1 = 4. A slope of 3/4 means rise = 3. So y = 2 + 3 = **5**.
5.7	The slope of the line through (2, 1) and (x, 7) is 2. Find x. Type a whole number.	**5**	Rise = 6. Slope 2 = 6 ÷ run, so run = 3. x = 2 + 3 = **5**.
5.8	Which pair of points makes a line with an undefined slope?	**(4, 1) and (4, 9)**	Both points have x = 4, so the run is 0 and the slope is **undefined**.

🔧 Error Depot — Error Detective

#	QUESTION	ANSWER	WHY
6.1	Piper counted a rise of 6 and a run of 4, then wrote the slope as 4/6. What went wrong?	**He flipped it. Slope is rise ÷ run, so it should be 6/4 = 3/2**	Slope = **rise ÷ run** = 6/4 = 3/2. Piper put the run on top.
6.2	Flurry found the slope through (1, 5) and (4, 2) like this: (5 - 2) ÷ (4 - 1) = 1. That's wrong! What is the correct …	**-1**	(2 - 5) ÷ (4 - 1) = -3 ÷ 3 = **-1**. Flurry subtracted y in the wrong order.
6.3	Mina says: "A vertical line has a slope of 0." What is the right correction?	**A vertical line's slope is undefined. A horizontal line's slope is 0.**	Vertical → run 0 → undefined. Horizontal → rise 0 → slope 0.
6.4	Flurry wrote the slope as 6/9. Her teacher says "Simplify it!" What should she write?	**2/3**	6 ÷ 3 = 2 and 9 ÷ 3 = 3, so 6/9 = **2/3**.
6.5	Looking at the table, Piper wrote the slope as 9/2 (using just the first column). What's the mistake? **X** 2 4 **Y** 9 13	**Slope uses the change between two columns: (13 - 9) ÷ (4 - 2) = 2**	Slope is a **change** ratio: (13 - 9) ÷ (4 - 2) = 4/2 = **2**.
6.6	A line goes down from left to right, but Piper wrote its slope as 3/2. What's the mistake?	**A downhill line has a negative slope, so it should be -3/2**	Going down means a negative rise, so the slope should be **-3/2**.

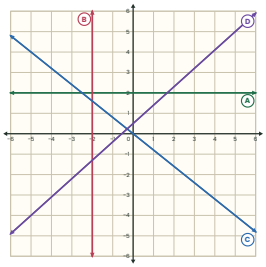
🎛️ Bonus: Master Controls — Master Challenge

#	QUESTION	ANSWER	WHY

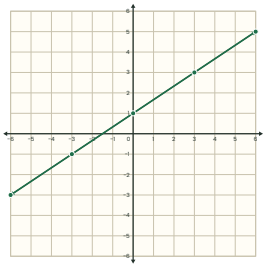
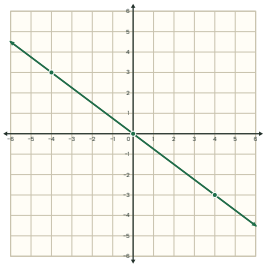
7.1	A wheelchair ramp rises 2 ft over a horizontal distance of 24 ft. What is the slope of the ramp? Simplified fraction.	**1/12**	2/24 = **1/12**.
7.2	Find the slope of the line through (-4, 9) and (6, -1). Whole number or simplified fraction (or undefined).	**-1**	Rise = $y_2 - y_1 = (-1) - 9 = -10$. Run = $x_2 - x_1 = 6 - (-4) = 10$. Slope = -10/10 = **-1** (simplified).
7.3	A line passes through (1, 3) and has slope 2. What is y when x = 4? Type a whole number.	**9**	Run = 3, rise = 2 × 3 = 6, so y = 3 + 6 = **9**.
7.4	A drone drops from 80 m to 20 m in 15 seconds. What is its rate of change in meters per second?	**-4 m per s**	(20 - 80) ÷ 15 = -60 ÷ 15 = **-4** m per second.
7.5	Which pair of points makes a line with a slope of 0?	**(-2, 5) and (7, 5)**	Both points have y = 5, so the rise is 0 and the slope is **0**.

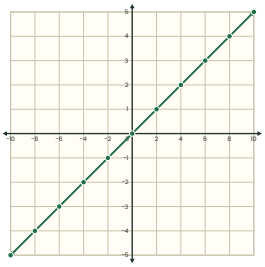
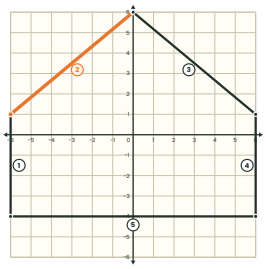
Weather Balloon Roundup – Quiz 1

🏁 Check 1 — Words & Types

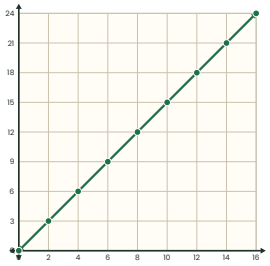
#	QUESTION	ANSWER	WHY
1.1	The horizontal change between two points on a line is called the _____.	**Run**	The horizontal change is the **run**.
1.2	Slope is another name for the _____ of change.	**rate**	Slope = **rate** of change = rise ÷ run.
1.3	What type of slope does each line have? Read left to right. 	**Line A: Zero; Line B: Undefined; Line C: Negative; Line D: Positive**	Positive lines climb, negative lines slide down, horizontal lines are zero, vertical lines are undefined.
1.4	Which description has a negative slope?	**A zip line sloping down from the tower**	A zip line that goes down as you move right has a **negative** slope.

📡 Check 2 — Slopes of Lines

#	QUESTION	ANSWER	WHY
2.1	Find the slope of the line. Whole number or simplified fraction. 	**2/3**	On the graph: Rise = $y_2 - y_1 = 3 - (-1) = 4$. Run = $x_2 - x_1 = 3 - (-3) = 6$. Slope = 4/6 = **2/3** (simplified).
2.2	Find the slope of the line. Whole number or simplified fraction. 	**-3/4**	On the graph: Rise = $y_2 - y_1 = (-3) - 3 = -6$. Run = $x_2 - x_1 = 4 - (-4) = 8$. Slope = -6/8 = **-3/4** (simplified).

2.3	Find the slope of the line. Careful: each square across is worth 2. Whole number or simplified fraction. 	**1/2**	On the graph: Rise = $y_2 - y_1 = 3 - (-3) = 6$. Run = $x_2 - x_1 = 6 - (-6) = 12$. Slope = 6/12 = **1/2** (simplified).
2.4	Find the slope of segment 2 (the thick orange one). Whole number, simplified fraction, or undefined. 	**5/6**	Segment 2: Rise = $y_2 - y_1 = 6 - 1 = 5$. Run = $x_2 - x_1 = 0 - (-6) = 6$. Slope = rise ÷ run = **5/6**.

🎈 Check 3 — Steeper, Flatter & Plot

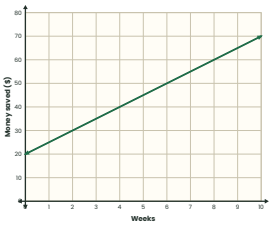
#	QUESTION	ANSWER	WHY
3.1	Which slope is steeper: 3/5 or 2/3?	**2/3**	3/5 = 9/15 and 2/3 = 10/15, so **2/3** is steeper.
3.2	Tap the slopes from steepest to flattest.	**5/2 → 2 → 1/2 → 1/3**	5/2 > 2 > 1/2 > 1/3.
3.3	The line passes through the red point (-2, 3) and has slope -2. Tap a second point on the line.	**(-1, 1) or (-3, 5)**	Slope -2 = rise -2 ÷ run 1. From (-2, 3) go 1 right and 2 down to **(-1, 1)** (or the other way: **(-3, 5)**).
3.4	Find the slope of the line. Golden gate! Careful: each square across is 2, each square up is 3. Whole number or simpl… 	**3/2**	On the graph: Rise = $y_2 - y_1 = 15 - 3 = 12$. Run = $x_2 - x_1 = 10 - 2 = 8$. Slope = 12/8 = **3/2** (simplified).

Slalom Run – Quiz 2

🚩 Gate 1 — Two Points

#	QUESTION	ANSWER	WHY
1.1	To find the slope through (1, 2) and (4, 8), which calculation is correct?	**(8 – 2) ÷ (4 – 1)**	Slope = $(y_2 - y_1) \div (x_2 - x_1)$ = (8 - 2) ÷ (4 - 1).
1.2	Find the slope of the line through (-3, -2) and (5, 2). Whole number or simplified fraction (or undefined).	**1/2**	Rise = $y_2 - y_1$ = 2 - (-2) = 4. Run = $x_2 - x_1$ = 5 - (-3) = 8. Slope = 4/8 = **1/2** (simplified).
1.3	Find the slope of the line through (4, 9) and (-2, -3). Whole number or simplified fraction (or undefined).	**2**	Rise = $y_2 - y_1$ = (-3) - 9 = -12. Run = $x_2 - x_1$ = (-2) - 4 = -6. Slope = -12/-6 = **2** (simplified).
1.4	Find the slope of the line through (-1, 7) and (5, -2). Whole number or simplified fraction (or undefined).	**-3/2**	Rise = $y_2 - y_1$ = (-2) - 7 = -9. Run = $x_2 - x_1$ = 5 - (-1) = 6. Slope = -9/6 = **-3/2** (simplified).

🏂 Gate 2 — Tables & Real-World Rates

#	QUESTION	ANSWER	WHY
2.1	Find the slope (rate of change) shown in the table. Whole number or simplified fraction. **X** 0 3 6 9 **Y** 5 3 1 -1	**-2/3**	Pick two rows, like (0, 5) and (3, 3). Change in y = 3 - 5 = -2. Change in x = 3 - 0 = 3. Slope = -2/3 = **-2/3**.
2.2	This table shows a constant rate of change. Find the missing y-value (the ?). **X** 1 3 5 7 **Y** 4 10 **?** 22	**16**	The rate is constant. Use two known columns to find it: (1, 4) and (3, 10) give slope 3. From x = 3 to x = 5, y changes by 3 × 2. So the missing y is **16**.
2.3	A weather balloon rises from 120 m to 480 m in 9 minutes. What is its rate of change in meters per minute?	**40 m per min**	(480 - 120) ÷ 9 = 360 ÷ 9 = **40** m per minute.
2.4	The graph shows money saved over time. What is the rate of change in dollars per week? 	**5 $ per week**	Rise = $y_2 - y_1$ = 50 - 20 = 30. Run = $x_2 - x_1$ = 6 - 0 = 6. Slope = 30/6 = **5** (simplified).

🏆 Gate 3 — Compare, Fix & Find

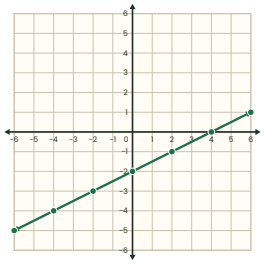
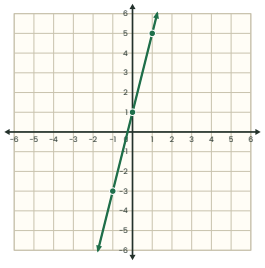
#	QUESTION	ANSWER	WHY
3.1	Runner A runs 14 km in 2 hours. Runner B's table is shown. Who is faster? **HOURS** 1 2 3 **KM** 8 16 24	**Runner B**	A: 14 ÷ 2 = 7 km/h. B: 8 km/h from the table. **Runner B** is faster.

3.2	What is the slope of the line x = 7?	**Undefined**	A vertical line has an **undefined** slope.
3.3	Mia found the slope through (2, 9) and (5, 3) as (9 - 3) ÷ (5 - 2) = 2. What went wrong?	**She subtracted in a different order: it should be (3 - 9) ÷ (5 - 2) = -2**	(3 - 9) ÷ (5 - 2) = -6 ÷ 3 = **-2**.
3.4	Golden gate! The slope of the line through (2, -1) and (6, y) is -3/2. Find y.	**-7**	Run = 4 = 2 × 2, so rise = -3 × 2 = -6. y = -1 + (-6) = **-7**.

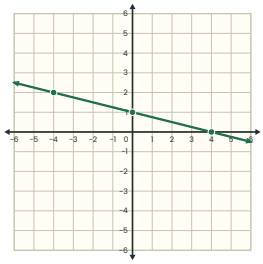
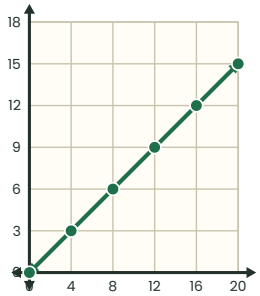
Homework – Secret Message

Message: F O L L O W T H E R I V E R → **FOLLOW THE RIVER** (hook for the next kit). Decoder table (answer → letter): -1/2 → A, 3/5 → D, -1/4 → E, 1/2 → F, 4/3 → G, 0 → H, 5 → I, -4 → K, 2/3 → L, -3 → O, -2/3 → Q, 5/3 → R, 2 → S, 4 → T, 3/4 → V, undefined → W, 3 → X, 3/2 → Z.
Trap letters (S, X, Z, Q, K, D, G, A) are common slips: flipped fraction or lost negative sign. If a student lands on one, ask them which slip they made.

Part A — the 14 locks

#	QUESTION	ANSWER	LETTER	WHY
1	Find the slope of the line. 	**1/2**	**F**	On the graph: Rise = $y_2 - y_1 = 0 - (-4) = 4$. Run = $x_2 - x_1 = 4 - (-4) = 8$. Slope = 4/8 = **1/2** (simplified).
2	Find the slope of the line through (-2, 7) and (1, -2).	**-3**	**O**	Rise = $y_2 - y_1 = (-2) - 7 = -9$. Run = $x_2 - x_1 = 1 - (-2) = 3$. Slope = -9/3 = **-3** (simplified).
3	Find the slope shown in the table.	**2/3**	**L**	Between two columns: change in y = 2, change in x = 3. Slope = 2/3 = 2/3.
4	A diver swims up 8 meters in 12 seconds. What is the rate of change in meters per second?	**2/3 m per s**	**L**	8 ÷ 12 = 8/12 = 2/3 meters per second.
5	A pile of snow melts from 24 cm tall to 9 cm tall in 5 hours. What is the rate of change in cm per hour?	**-3 cm per h**	**O**	(9 - 24) ÷ 5 = -15 ÷ 5 = -3 cm per hour (negative: the pile is shrinking).
6	Find the slope of the line through (4, 7) and (4, -1). (Write “undefined” if it has no slope.)	**undefined**	**W**	Both points have x = 4, so the run is 4 - 4 = 0. You can't divide by zero, so the slope is **undefined** (a vertical line).
7	Find the slope of the line. 	**4**	**T**	On the graph: Rise = $y_2 - y_1 = 5 - (-3) = 8$. Run = $x_2 - x_1 = 1 - (-1) = 2$. Slope = 8/2 = **4** (simplified).
8	What is the slope of the line y = -3?	**0**	**H**	y = -3 is a horizontal line, so the rise is 0 and the slope is 0.

X	0	3	6	9
Y	1	3	5	7

9	Find the slope of the line. 	**-1/4**	**E**	On the graph: Rise = $y_2 - y_1 = 0 - 2 = -2$. Run = $x_2 - x_1 = 4 - (-4) = 8$. Slope = -2/8 = **-1/4** (simplified).
10	Find the slope of the line through (-3, -4) and (3, 6).	**5/3**	**R**	Rise = $y_2 - y_1 = 6 - (-4) = 10$. Run = $x_2 - x_1 = 3 - (-3) = 6$. Slope = 10/6 = **5/3** (simplified).
11	A rocket climbs 150 meters in 30 seconds. What is the rate of change in meters per second?	**5 m per s**	**I**	150 ÷ 30 = 5 meters per second.
12	Find the slope of the line. Careful: x counts by 4s and y counts by 3s. 	**3/4**	**V**	From (4, 3) to (12, 9): rise = 6, run = 8, slope = 6/8 = 3/4.
13	Find the slope shown in the table. **X** 0 4 8 12 **Y** 9 8 7 6	**-1/4**	**E**	Between two columns: change in y = -1, change in x = 4. Slope = -1/4.
14	A tank fills with 20 liters of water in 12 minutes. What is the rate of change in liters per minute?	**5/3 L per min**	**R**	20 ÷ 12 = 20/12 = 5/3 liters per minute.

Part B — model answers

#	QUESTION	MODEL ANSWER	TUTOR NOTE
15	Name the type of slope for each line (positive, negative, zero, or undefined).	**A: negative, B: zero, C: undefined, D: positive**	Read left to right. A goes down (negative), B is flat (zero), C is vertical (undefined), D goes up (positive).
16	The graph shows a hiker's elevation over time. (a) Find the rate of change, with units. (b) Explain in a sentence what it means.	**(a) 150 meters per hour. (b) The hiker climbs 150 meters every hour.**	Points (0, 200) and (4, 800): rise 600, run 4, slope 150. Units: meters per hour.
17	Sam finds the slope through (2, 5) and (6, 1) like this: (5 - 1) ÷ (6 - 2) = 4/4 = 1. Find Sam's mistake and give the correct slope.	**Sam subtracted the y-values in a different order than the x-values. Correct: (1 - 5) ÷ (6 - 2) = -4/4 = -1.**	Both subtractions must use the same order.

18	Graph the line through (1, 4) with slope -2/3. Plot at least three points and draw the line. Show how you moved from the first point.	**Slope -2/3 = down 2 for every 3 to the right. Points on the line: (1, 4), (4, 2) [right 3, down 2], and (-2, 6) [left 3, up 2]. The line slants downward from left to right.**	
19	Car A's trip is in the table. Car B travels 240 miles in 4 hours. Which car is faster, and by how many miles per hour?	**Car A: 55 mph. Car B: 60 mph. Car B is faster by 5 mph.**	Find each rate, then subtract.
20	In your own words, explain the difference between a zero slope and an undefined slope. Give a real-world example of each.	**Zero slope: a flat, horizontal line (rise 0) — e.g., a flat lake path. Undefined slope: a straight up-and-down, vertical line (run 0, can't divide by zero) — e.g., a cliff wall.**	

Tutor Notes – common slips and what to say

What the student does	Why it happens	What to say
Writes run ÷ rise (4/6 instead of 6/4)	Reads the fraction in the order they counted it	"Rise is vertical and goes on top: *rise over run*." Count up/down first.
Forgets the negative on a downhill line	Counts the squares but not the direction	"Walk left → right. Did you go up or down?" Down = negative rise.
Mixes the order of subtraction	$(y_2 - y_1) \div (x_1 - x_2)$	"Same point first on top and bottom." Label point 1 and point 2.
Says a vertical line has slope 0 (or horizontal is undefined)	Zero and undefined feel alike	"Undefined = Up-and-down. Zero = flat like the floor." Test with rise ÷ run.
Reads squares instead of axis numbers	Scaled graphs (each square = 3, 5…)	"Look at the numbers on the axis, not the squares."
Does not simplify (6/9)	Thinks any correct fraction is fine	"Divide top and bottom by the same number until you can't." The games ask for it.
Uses y ÷ x from one table row as the slope	Confuses slope with unit rate through the origin	"Slope compares how much y *changes* to how much x *changes*."
Thinks -4 is flatter than 2	Compares signed numbers instead of steepness	"Sign = direction. Size = steepness."

Skill map (report card tips)

Skill	Tip shown to the student
Slope vocabulary	Re-read the Field Guide: rise = vertical change, run = horizontal change, slope = rise ÷ run.
Types of slope	Remember: uphill left→right = positive, downhill = negative, flat = zero, a wall = undefined.
Slope from a graph	Pick two corner points, count the rise (up/down) then the run (right), and simplify the fraction.
Slope of line segments	Treat each segment like its own line: find two end points, then rise ÷ run. Flat = 0, wall = undefined.

Comparing steepness	Bigger number (ignoring the sign) = steeper. Compare fractions by making the bottoms match.
Graphing from a slope	Start at the point, go RIGHT for the run, then UP (positive) or DOWN (negative) for the rise.
Slope from two points	slope = $(y_2 - y_1) \div (x_2 - x_1)$. Subtract in the SAME order on top and bottom.
Slope from a table	Compare two rows: change in y ÷ change in x. Check that every pair of rows gives the same rate.
Real-world rates	Rate of change = change in the dependent variable ÷ change in the independent variable. Keep the units!
Horizontal & vertical lines	y = number is flat (slope 0). x = number is a wall (slope undefined).
Missing coordinates	Write slope = rise ÷ run, fill in what you know, then solve for the missing number.
Error analysis	Check the order of subtraction, the sign, the rise-over-run flip, and whether the fraction is simplified.